June 2026
Vol. 102 No. 3

Routledge
Taylor & Francis Group

Economic Geography

ISSN: 0013-0095 (Print) 1944-8287 (Online) Journal homepage: www.tandfonline.com/journals/recg20

Integrating Geoeconomics into the GPN Framework: The Internationalization of Chinese EV Battery Firms amid China–EU Strategic Interaction

Huiwen Gong, Yue Guo, Yijia Chen & Bjørn Asheim

To cite this article: Huiwen Gong, Yue Guo, Yijia Chen & Bjørn Asheim (2026) Integrating Geoeconomics into the GPN Framework: The Internationalization of Chinese EV Battery Firms amid China–EU Strategic Interaction, Economic Geography, 102:3, 179-208, DOI: 10.1080/00130095.2026.2630648

To link to this article: https://doi.org/10.1080/00130095.2026.2630648

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group, on behalf of Clark University.

View supplementary material

Published online: 10 Mar 2026.

Submit your article to this journal

Article views: 6348

View related articles

View Crossmark data

Citing articles: 6 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=recg20

ECONOMIC GEOGRAPHY 179 102(3):179–208. www.economicgeography.org

Integrating Geoeconomics into the GPN Framework: The Internationalization of Chinese EV Battery Firms amid China–EU Strategic Interaction

Huiwen Gong
Centre for Innovation Research
School of Business and Law
University of Stavanger
Norway
huiwen.gong@uis.no
and
CIRCLE (Centre for Innovation Research)
Lund University
Lund
Sweden

Yue Guo
School of International Relations
University of International Business and Economics
China
guoyue_econgeo@163.com

Yijia Chen
School of Economics and Management
Dongguan University of Technology
Dongguan
China
yjchenrenaissance@gmail.com

Bjørn Asheim
Centre for Innovation Research
School of Business and Law
University of Stavanger
Stavanger
Norway
and
CIRCLE (Centre for Innovation Research)
Lund University
Lund
Sweden
bjorn.asheim@circle.lu.se

abstract

Over the last two decades, strategic emerging global production networks (GPNs) have reshaped advanced manufacturing, challenging the dominance of developed economies. Yet existing GPN theories pay limited attention to the geoeconomic logics driving these transformations. This article develops a conceptual–analytical framework that integrates geoeconomics into GPN analysis, viewing states as geoeconomically competing and strategically selective actors and firms as active agents navigating these dynamics. Focusing on the electric vehicle battery industry, it examines how leading Chinese firms expand into Europe amid the strategic interaction between Chinese and EU agendas, each deploying distinct geoeconomic instruments. The analysis shows how corporate competitive imperatives (cost-capability optimization, market access, and financial discipline) are mediated by geopolitical and regulatory pressures, prompting varied firm coping strategies. The study advances understanding of globalization under hybrid state–firm logics and highlights the rise of emerging GPNs led by non-Western firms in an era of intensified geoeconomic competition.

This article has been corrected with minor changes. These changes do not impact the academic content of the article.

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group, on behalf of Clark University. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

Key Words:
emerging-market lead firms
electric vehicle (EV) battery industry
geoeconomics
global production networks (GPNs)
hybrid state–firm driven GPN

Acknowledgments

We are grateful to Editor Siobhan McGrath and four anonymous reviewers for their constructive comments on earlier versions of this article. Huiwen Gong acknowledges financial support from the Norwegian Research Council, Grant 352805, and EU Marie-Curie Standard Individual Fellowship, Grant 894063. She thanks all the interviewees for sharing their knowledge and expertise. Yue Guo acknowledges financial support from the National Natural Science Foundation of China, Grant 42501212, and the Humanities and Social Science Fund of Ministry of Education of China, Grant 25YJCGJW002. The corresponding author Yijia Chen acknowledges the support from the Start-up Fund of Dongguan University of Technology, Grant 221110246. We thank Dr. Ruilin Yang for sharing the FDI data source.

Over the past two decades, the emergence of global production networks (GPNs) in advanced manufacturing, such as cleantech, telecommunications (5G), and semiconductors, has marked a notable shift from the traditional dominance of developed nations (Bridge and Faigen 2022; Coe and Gibson 2023; Aoyama, Song, and Wang 2024; Li and Gong 2025). Firms from emerging economies have taken on increasingly pivotal roles in shaping these industries, signaling a transformation in global industrial dynamics (Horner and Nadvi 2018; IEA 2021). This power shift, coupled with rising geopolitical tensions and recurring crises, is fueling the reconfiguration of strategic GPNs (Gong et al. 2022). Despite its profound impact on the global political economy, existing GPN theories fall short of capturing the complexities of this transition. For instance, while studies have illuminated power dynamics and causal mechanisms within established production networks (Henderson et al. 2002; Coe and Yeung 2015), they struggle to address the nuances of newer strategic GPNs led by firms from emerging economies (Horner and Nadvi 2018; Umar and Yang 2025): How can we better understand the motivations and instruments employed by states in the context of contemporary global competition in strategic sectors, and how do these shape the reconfiguration of strategic emerging GPNs? Furthermore, how do emerging-market multinationals (EMNEs) navigate the tensions between geopolitical imperatives and commercial interests as they take on leading roles in these emerging GPNs? Finally, how are dynamic imperatives such as cost-capability optimization, market imperative, and financial discipline, being reconfigured under conditions of intensified geoeconomic competition, compared to how they operated during the era of neoliberal globalization?

This article addresses these gaps by proposing a conceptual-analytical framework that brings key insights from geoeconomics to GPN research for understanding the ongoing hybrid state–firm driven dynamics in emerging strategic GPNs. We argue for the importance of disentangling the mechanisms underpinning firm and state behavior, an essential analytical move in an era of intensifying geopolitical tensions. Using the electric vehicle (EV) battery industry as a case study—a strategic sector at the heart

Supplemental data for this article can be accessed online at https://doi.org/10.1080/00130095.2026.2630648.

of the global energy transition and increasingly shaped by Chinese leadership, we examine how Chinese firms navigate geopolitical uncertainties while pursuing commercial interests in Europe. We focus on the interplay between the agendas of China and the EU and the geoeconomic instruments structuring these dynamics. In addition, we analyze how the competitive dynamic forces outlined in GPN 2.0, that is, cost-capability ratio, market imperative, and financial discipline (Coe and Yeung 2015), are reshaped under conditions of geoeconomic competition, contrasting with their operation during the era of neoliberal globalization.

Understanding Global Production and Political Economy: Key Theories, Insights, and Challenges

Contemporary Transformations in the Global Political Economy

Two major shifts are reshaping the global political economy and challenging existing analytical tools. First, since the mid-2010s, rising geopolitical tensions, technological rivalries, and successive global crises have disrupted established GPNs and spurred the formation of new ones—particularly in strategic sectors such as net-zero technologies, semiconductors, and telecommunications (Bridge and Faigen 2022; Coe and Gibson 2023; Aoyama, Song, and Wang 2024; Li and Gong 2025). The liberal trade logic that once underpinned globalization is gradually giving way to a more state-centered form of economic organization, marked by national security concerns and industrial policy intervention (Herranz-Surrallés, Damro, and Eckert 2024). Cooperation and coercion measures now coexist in the reorganization of supply chains and global market access. This marks a radical shift from firm-led globalization to hybrid state–firm driven GPN dynamics, where states and corporations jointly shape production networks through intertwined strategic, political, and economic logics (Li and Gong 2025). Production strategies are increasingly guided by geopolitical alignment and risk mitigation, resulting in a regionally structured, multipolar world economy.

Second, EMNEs, notably from China, are increasingly taking leading roles in some strategic sectors (Langguth 2024; Li and Gong 2025; Umar and Yang 2025). These firms have advanced beyond low-value functions to become dominant actors in areas such as batteries, solar power, and EVs (Lema, Fu, and Rabellotti 2020; Gong and Hansen 2023). In the case of China, their rise has been supported by long-term state planning, strategic industrial policy, integration into global innovation systems, and the ability of Chinese firms to capitalize on early-stage green windows of opportunity (Binz et al. 2020). In the case of the EV battery sector, Chinese firms now control midstream manufacturing and materials processing, where Western countries remain relatively weak (Bridge and Faigen 2022). In response, the EU and others have launched green industrial policies to reduce dependency and localize value capture (Meckling 2025). Yet countries in South America and Africa often remain trapped in extractive roles, reproducing global inequalities (Bridge and Faigen 2022). At the same time, a few Global South economies illustrate how some peripheral regions can benefit from geopolitical tensions by leveraging regulatory fragmentation and great-power rivalry as strategic assets in the (re)organization of production (Umar and Yang 2025).

These two shifts together create tremendous uncertainty for EMNEs' internationalization processes. Policy discourses, like *decoupling*, *de-risking*, and *technology sovereignty*, reflect how firms are increasingly viewed as strategic or even threatening geopolitical entities (Edler et al. 2023; Teixeira 2024a). While some of these firms have secured global leadership, the ways they manage the tension between global expansion and growing geoeconomic restrictions remain poorly understood. The pathways through which these firms rise and the reciprocal strategies between firms and states, remain undertheorized. To analyze these developments, two key approaches offer important and complementary insight: the GPN framework and the geoeconomic approaches.

Global Production Networks and Geoeconomics

The GPN 1.0 framework, introduced in the early 2000s, mapped the complex networks of firms and territorial institutions shaping increasingly globalized production through three key elements—value, power, and embeddedness (Henderson et al. 2002). GPN 2.0 (mid-2010s) advanced a more dynamic, multidimensional approach, emphasizing how cost-capability ratios, market imperatives, financial discipline, and risk environments shape production networks (Coe and Yeung 2015). More recent work further expands its analytical reach by engaging with domains such as the state, finance, labor, environment, materials, and development (Coe and Yeung 2019; Hess and Horner 2024; Coe et al. 2025). Yet, much of this scholarship was developed during an era of neoliberal globalization, presuming Western lead firms operating in relatively open markets supported by liberal trade regimes and cooperative state behavior (Henderson and Hooper 2021). While the role of the state has received increasing attention (Smith 2015; Horner 2017; Mayer and Phillips 2017), previous work often relies upon the Weberian conceit of a state–market divide (Glassmann 2011) or typologies of state roles (Horner 2017) to make their arguments (Werner 2021). While GPN scholars have also drawn on neo-Gramscian and/or strategic-relational approach (Teixeira 2024c; Vezzoni 2024; Coe et al. 2025), these traditions are rarely integrated into a coherent framework. This conceptual fragmentation limits GPN's ability to account for

contemporary state behavior, especially in contexts where states actively disrupt, steer, and restructure production networks (for recent contributions, see Herranz-Surrallés, Damro, and Eckert 2024; Hess and Horner 2024; Pavlínek 2024). EMNEs today are therefore embedded in contexts marked by strong state visions and geostrategic intent (Horner and Nadvi 2018; Li and Gong 2025). Both the conceptualization of the state and assumptions about firm agency in GPN studies therefore require significant updating.

Geoeconomics provides a useful lens for addressing these limitations and understanding hybrid state–firm GPN dynamics. Over the past three decades, geoeconomics has gained increasing traction in international political economy and political geography (Sparke 1998; Mallin and Sidaway 2024), driven by a growing recognition that states strategically deploy economic tools, such as trade policy, investment screening, and industrial subsidies, to pursue geopolitical goals. While geoeconomic behavior is not new, with historic precedents in the interwar period and the 1970s, the current phase is marked by intensified state intervention, greater institutional capacity, and deeper integration into global markets (Blackwill and Harris 2016). Today's strategies target critical technologies, infrastructures, and resources, increasingly linking industrial policy to security and climate agendas under conditions of multipolar rivalry (Aoyama, Song, and Wang 2024; Li and Gong 2025).

Unlike the cumulative GPN tradition, geoeconomics remains a rather fragmented field (Glassman 2025). Scholars continue to debate whether it should be understood as a theory, a discourse, or a strategic practice, and its meaning often shifts across geopolitical and disciplinary contexts (Sparke 2018; Mallin and Sidaway 2024). Rather than treating geoeconomics as a unified theory, we adopt it here as a critical analytical lens to foreground the strategic use of economic instruments by states in pursuit of geopolitical or geostrategic objectives. Crucially, the value of geoeconomics lies in helping better theorize how state–firm alliances, often formed in non-Western institutional contexts (Alami and Dixon 2024), actively reshape strategic GPNs. This is especially important given the rise of EMNEs operating under political–economic logics distinct from those in liberal market economies. While GPN research has long emphasized firm agency and interfirm coordination, geoeconomics draws attention to how long-term strategic capital flows, guided by coherent state visions, reorganize global value creation and capture.

In sum, GPNs provide a relational and spatial vocabulary centered on lead firms and network configurations, typically informed by a functionalist understanding of the multiple roles of the state. Geoeconomics, by contrast, foregrounds the strategic rationalities, coercive capacities, and geopolitical logics driving state intervention. Combining these perspectives allows us to better analyze hybrid state–firm formations and politically mediated value chains that increasingly define contemporary global production. The next section proposes a conceptual–analytical framework for integrating these insights.

Framework Development: Bringing Geoeconomics in Conversation with GPN Research

The framework aims to bring insights from geoeconomics to the GPN literature for understanding the co-constitutive relationship between state strategies and firm responses in shaping GPN trajectories. Our vantage point is strategic emerging GPNs led by

EMNEs, particularly from China. Existing GPN studies largely center on lead firms from advanced industrialized countries (e.g., the US, the UK, Germany, Japan) known for innovation capacity, brand power, and organizational control. In contrast, we focus on non-conventional lead firms rising in strategic sectors but operating under distinct political–economic logics, marked by close state ties, long-term capital commitment, and geopolitical ambition. Figure 1 presents a three-layered framework for analyzing geoeconomic

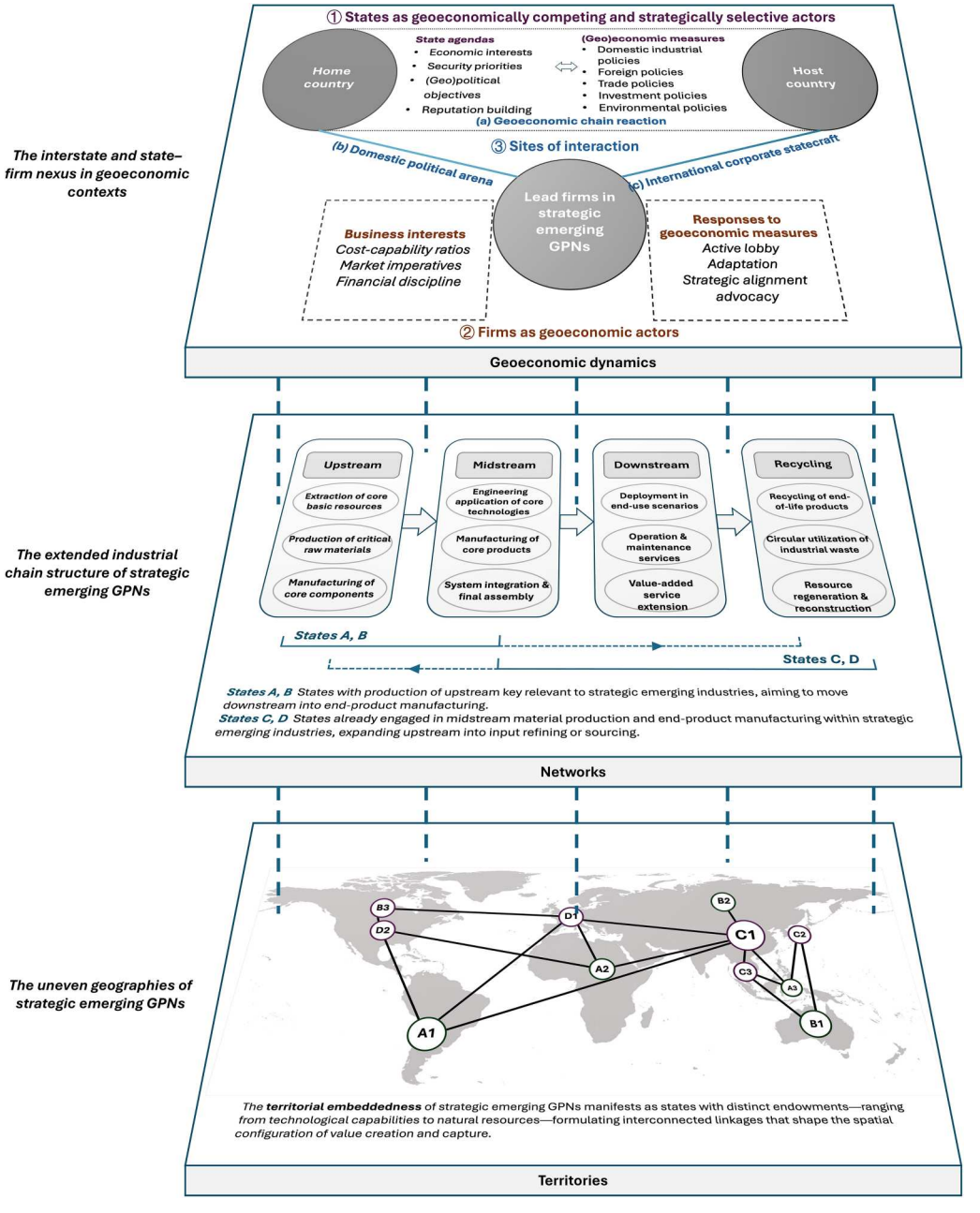


Figure 1. Conceptual–analytical framework.

dynamics in strategic emerging GPNs, including (1) territories, (2) networks, and (3) geoeconomic dynamics. While interrelated, each layer captures a distinct dimension of GPN transformation.

The networks layer traces the industrial chain from upstream resource extraction to downstream deployment and recycling, showing how value is created and coordinated across stages and geographies (Coe and Gibson 2023; Coe et al. 2025). The territories layer highlights spatial embeddedness, illustrating how states with different endowments, such as technological capabilities or natural resources, are linked through evolving value chains. Together, these two layers show how industrial organization and territorial conditions jointly shape GPN configurations (Bridge and Faigen 2022).

The geoeconomic dynamics layer constitutes the core conceptual contribution of the framework, foregrounding the co-constitutive relationship between state strategies and firm responses in shaping GPN trajectories. This layer comprises three key elements. First, states are conceptualized as geoeconomically competing and strategically selective actors that pursue distinct agendas through the strategic selectivity of a range of geoeconomic instruments (see ① in Figure 1 and elaboration in "States as Geoeconomically Competing and Strategically Selective Actors"). Second, firms are understood as active agents in geoeconomic processes, driven by business interests yet capable of responding to state-imposed geoeconomic measures in varied ways (see ② in Figure 1 and elaboration in "Firms as Geoeconomic Actors"). Third, the sites of interaction encompass both state–firm relations and interstate dynamics. These sites serve as arenas where competing interests are negotiated, alliances are formed, and conflicts may emerge, shaping the trajectories of strategic emerging GPNs (see ③ in Figure 1 and elaboration in "Sites of Interaction"). Overall, this framework contributes to a more nuanced understanding of globalization under hybrid state–firm logics, highlighting how contemporary global economic integration is increasingly driven by the interplay between state geoeconomic agendas and corporate strategic responses. The remainder of this section details the mechanisms through which their interactions influence GPNs.

States as Geoeconomically Competing and Strategically Selective Actors

The first element of our framework views states as *geoeconomically competing* and *strategically selective* actors within GPNs, drawing from geoeconomics and critical state theories. The first aspect captures how states pursue distinct economic, geopolitical, and symbolic projects to secure leadership in the global political economy (Cox 1983), while the second underscores that the realization of these projects depends on the selective deployment of diverse geoeconomic instruments across various policy domains (Jessop 2001). This echoes Glassman (2011), Werner (2021), and Vezzoni (2024), who argue that understanding states in GPNs requires moving beyond *what* states do (e.g., typology development), to *why* states act as they do (e.g., the geoeconomic agendas driving their actions), and *how* they act (i.e., the multifaceted geoeconomic instruments they employ). This reconceptualization resonates with recent calls by Hess and Horner (2024) and Coe et al. (2025) for greater engagement with neo-Gramscian and strategic-relational approach in understanding states in GPNs.

To interpret *why* states maintain or disrupt strategic GPNs, we draw on the neo-Gramscian notion of the state as an agent of geoeconomic competition, engaged in

struggles for leadership and influence within the global political economy (Cox 1983). States articulate strategic projects that combine economic, political, security, and symbolic objectives aimed at consolidating or advancing their position in global competition. Economically, states aim to enhance industrial competitiveness, foster innovation, and secure growth through policies that attract investment, support strategic sectors, and strengthen their position in GPNs (Mayer and Phillips 2017). Politically and geopolitically, states prioritize technological sovereignty, national security, or maintaining extraterritorial influence when making decisions about GPN participation (Hess and Horner 2024; Kalvelage and Tups 2024; Selwyn, Bernhold, and Leyden 2025). Additionally, states may pursue symbolic power by positioning themselves as leaders in emerging GPNs and offering affordable or normatively appealing products to global markets (Li 2020). Collectively, these projects reveal the geoeconomically competing nature of states: Their actions are not merely functional responses to market dynamics but components of broader competitions over control of key sectors and value chains.

The realization of these projects depends on state interventions, that is, its capacity to advance geoeconomic competition through the targeted use of specific policy instruments (the *how* question). From the perspective of the strategic-relational approach, these measures are best understood as *strategically selective* state interventions (Jessop 2001), conditioned by historically and institutionally embedded configurations of power (Blackwill and Harris 2016). States can draw upon a wide range of instruments to advance their geostrategic goals (Curran 2015; Herranz-Surrallés, Damro, and Eckert 2024), which we call geoeconomic measures. We define such measures as instruments (legally binding or nonbinding) whose design or application explicitly serves geopolitical or geostrategic objectives (e.g., measures that alter international production structures or reduce dependence on rival powers). This definition distinguishes geoeconomic measures from general economic or regulatory policies, which may affect global production indirectly (e.g., domestic labor regulation or general environmental standards) but are not primarily deployed for strategic positioning. Hence, not all state interventions qualify as geoeconomic measures. They *only* acquire a geoeconomic character when deliberately used to shape cross-border competition or control access to critical technologies, markets, or resources. For instance, *domestic industrial policies* only become geoeconomic measures if they are crafted to ensure technological sovereignty and safeguard territorial–industrial integrity, especially in sectors where reliance on profit-oriented firms (often driven by short-term shareholder value) may conflict with long-term national interests (Edler et al. 2023). Moreover, the weaponization of interdependence exemplifies how global economic networks shape state coercion (Farrell and Newman 2019; Aoyama, Song, and Wang 2024). Coercive tools, such as economic sanctions and embargoes, often framed as instruments of *foreign policies*, are increasingly deployed to advance broader geoeconomic objectives, restricting rivals' access to markets, technologies, or financial systems while targeting firms, sectors, or entire countries deemed critical to the sanctioning state's competitive positioning (Pavlínek 2024). Likewise, *trade policies* and *investment policies* can serve as selective tools of geoeconomic competition, as seen in how antisubsidy and antidumping measures, tariffs, and investment screening mechanisms are used to defend national industries from foreign domination and to reshape the geography of production in line with state interests (Alami 2024). Additionally, *environmental policies*, once peripheral to

geoeconomic agendas, have become increasingly politicized and employed as green trade barriers, functioning as instruments of regulatory geoeconomics (Herranz-Surrallés, Damro, and Eckert 2024).

Firms as Geoeconomic Actors

The second element of our framework relates to understanding firms as geoeconomic actors (Eckert 2024; Lim 2025), driven by business imperatives yet capable of responding to state-imposed geoeconomic measures in varied ways. While GPN 2.0 has offered valuable insights into firm agency through its theorization of intrafirm coordination, interfirm control, interfirm partnership, and extrafirm bargaining (Coe and Yeung 2015), these strategies are grounded in a context of neoliberal globalization. Empirical evidence remains limited in capturing how firms react to politicized forms of market access and state-led (de)coupling efforts (for exceptions, see Blažek and Lypianin 2024; Pavlínek 2024; Klagge et al. 2025). While GPN work has started to acknowledge the increasing importance of geopolitics and state agendas (e.g., Glassman 2011; Smith 2015; Yeung 2023; Hess and Horner 2024; Selwyn, Bernhold, and Leyden 2025), few studies have systematically theorized how firms actively navigate and respond to such geoeconomic constraints and opportunities (Pavlínek 2024). To address this, we introduce a typology of four corporate geoeconomic strategies that foreground how firms engage with geoeconomic risks and regulatory fragmentation while still pursuing commercial interests.

First, firms may engage in *active lobbying and selective noncompliance* when responding to proposed, nonbinding geoeconomic measures. This strategy targets measures that already exist in draft, advisory, or soft-law form. Such actions typically involve attempts to influence or delay the implementation of measures that are advisory, contested, or not yet fully enforceable. Firms may voice opposition through soft resistance (e.g., public statements, lobbying, or negotiation), or through limited operational responses such as delaying investment or threatening to withdraw from projects (Blažek and Lypianin 2024). These responses are strategic and discursive rather than defiant: Even when firms disagree, compliance is required for binding measures, and therefore this category applies primarily to nonmandatory instruments such as state recommendations or proposed policy drafts. Where measures are compulsory, disagreement may still manifest rhetorically but not in operational noncompliance (Pavlínek 2024). Thus, this category primarily concerns *existing but nonmandatory* instruments. Second, firms adopt *compliance and adaptation* with mandatory measures when confronted with binding state-imposed regulations or trade rules that determine market access. In these situations, firms must comply to preserve their operations but adjust their organizational and market strategies to minimize costs and maintain competitiveness. Adaptation may involve restructuring supply chains, altering ownership or partnership arrangements, or reconfiguring production to align with divergent geopolitical jurisdictions. A well-documented example is the *one company, two systems* approach (Choer Moraes and Wigell 2020) in which firms tailor their governance models and market activities to meet distinct regulatory and political requirements. Third, firms may pursue *strategic alignment* with state agendas when geoeconomic measures or policies coincide with their commercial objectives. This alignment extends beyond passive acceptance to active collaboration with state actors to secure resources, legitimacy, or market access (Liu and

Dicken 2006). Finally, firms may engage in *advocacy and policy shaping* strategies to influence future regulatory frameworks or industrial policies that have not yet been formulated, finalized, or enacted. Unlike the first strategy, which targets existing drafts, this form of engagement seeks to shape the direction, priorities, and content of new policy initiatives before they enter the formal policy-making process. These efforts typically occur through industry associations, business councils, and consultations with public authorities, where firms collectively seek to shape the institutional environment in ways that reduce uncertainty or create new market opportunities. These strategies are not mutually exclusive: Firms may combine them across scales and over time, reflecting the multiscalar, dynamic nature of geoeconomic governance and corporate strategy in GPNs.

Importantly, these geoeconomic roles should not obscure firms' core identities as economic actors. Their priorities (profit maximization, shareholder value, and market expansion) are central, though shaped by ownership models. In this light, GPN 2.0's three competitive dynamics, that is, cost-capability ratios, market imperatives, and financial discipline, remain analytically useful. However, our contribution lies in showing how firms interpret and operationalize these dynamics differently under geoeconomic pressure. Our empirical analysis illuminates how these logics are recalibrated in response to strategic state agendas, policy constraints, and geopolitical rivalry.

Sites of Interaction

The third element of our framework concerns sites of interaction, encompassing both interstate dynamics and state–firm relations. In the realm of interstate dynamics, the agendas and instruments of one state often provoke countermeasures by others, triggering what Choer Moraes and Wigell (2020) term a *geoeconomic chain reaction*—a sequence of strategic moves and countermoves as states seek to reposition themselves in response to shifting configurations of power and production. From a GPN perspective, these dynamics carry profound implications for how GPNs are governed, contested, and reconfigured. They reflect a form of relational statecraft, in which the geoeconomic chain reaction becomes a key mechanism through which the spatial and institutional architecture of GPNs is transformed, compelling firms to recalibrate their global strategies in response to escalating and strategically selective interventions by multiple states (Blackwill and Harris 2016; Bridge and Faigen 2022).

Equally important are state–firm relations. We use the term *domestic political arena* to describe how firms interact with their home state[1] and formulate strategies in response to state-led geoeconomic policies. Conversely, *international corporate statecraft* refers to how firms engage with and respond to geoeconomic measures imposed by foreign states in the context of their global operations. In both cases, firms may adopt a range of strategic responses depending on the configuration of power relations and interest alignments between state and firm. The state–firm nexus thus emerges as a dynamic, historically embedded mechanism (Guo et al. 2023; Teixeira 2024b), through which states

[1] Often in a structure that encompasses national–regional–local dimensions. However, for the sake of simplification, we focus primarily at the national level in this article.

and firms construct institutional, policy-coordinated, and interest-convergent linkages that underpin firm strategies in both domestic and international arenas.

Analytically, these sites of interaction can be examined independently or integrated with the earlier two dimensions of analysis. Regardless of the approach, it is essential to attend closely to the power dynamics and reciprocal adjustments that shape these interactions.

Case Justification and Research Methods

In recent years, Chinese EV battery firms have rapidly expanded in Europe. In 2022, automotive investments accounted for 53 percent of all Chinese investments in Europe (the EU + UK), a 33 percent rise over 2021 (Guo et al. 2024). Moreover, in 2023, over two-thirds of Chinese foreign direct investment (FDI) was made in the EV sector, up from 41 percent in 2022 (Sebastian, Goujon, and Meyer 2024). Taking greenfield FDI, one of the major investment forms, as an example (see Figure 2a), Chinese EV-related greenfield investment in Europe has increased significantly since 2018, reaching a peak of over US$9 billion in 2022—the highest level recorded in the past decade. Although investment volumes declined slightly in 2023, they remained substantial, roughly twice the level observed in 2020. Geographically, Hungary has emerged as a central hub for Chinese EV battery manufacturing investment (see Figure 2b). In addition, Germany, Sweden, Slovakia, the UK, and Belgium have served as key destinations, reflecting their strategic importance in the evolving European EV battery supply chain. In terms of functional distribution, Chinese EV-related greenfield investment over the period 2013–24 concentrated mainly in manufacturing (approximately 95

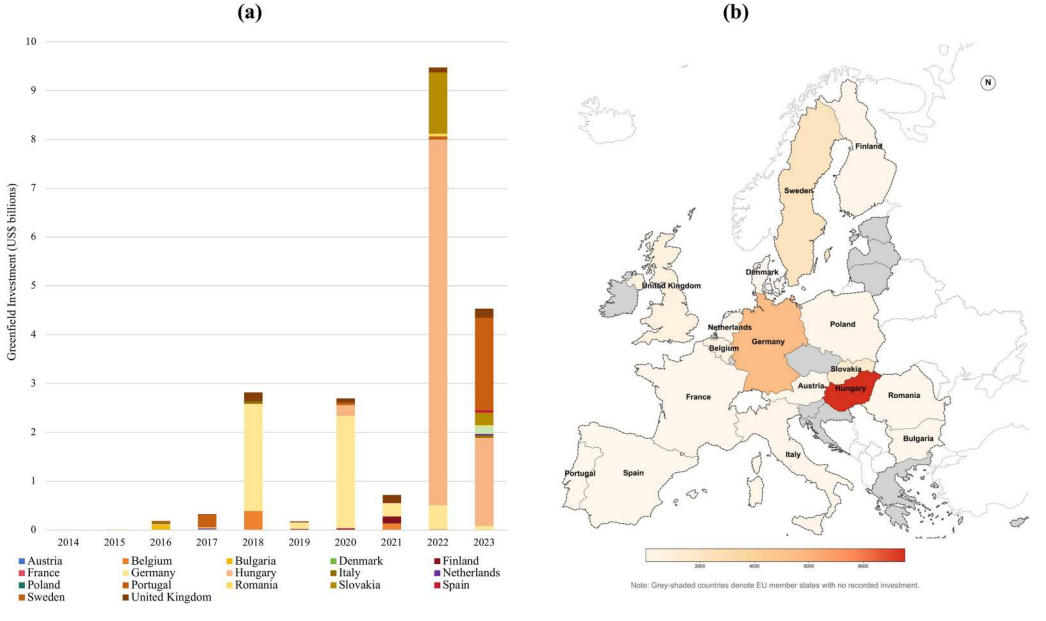


Figure 2. Year–country (a) and cumulative distribution (b) of Chinese EV greenfield investment in Europe over the period 2014–23 (billion US dollars).
Source: FDI markets; authors' calculation.

percent, see Figure 3). Investments in other functions, such as research and development (R&D) (approximately 3.2 percent), headquarters (approximately 1.5 percent), and sales (approximately 0.2 percent), were comparatively limited. Moreover, a closer look at the value chain composition (see Figure 4) suggests a shift toward greater diversification. While midstream battery manufacturing still dominates, recent years have witnessed growing investment in both upstream (materials and components) and downstream (vehicle manufacturing) segments. Especially, downstream investment in 2023 exceeded that of the midstream for the first time. This trend points to the gradual formation of a more localized whole-vehicle production by Chinese firms within Europe. In terms of ownership structure, most Chinese EV battery firms operating in the EU exhibit a mixed ownership model (see Appendix A in the online material for details). While private ownership is more common among battery firms, state ownership is more frequently observed in automakers.

This research draws on sixty-four semistructured interviews conducted between 2021 and 2024 across Europe and China. Interviewees were purposefully selected to capture diverse perspectives along the EV battery value chain, including senior managers and strategy directors from major Chinese and European firms ($n = 26$), scientists and technical experts ($n = 21$), officials from national and EU-level agencies ($n = 8$), and representatives of industry associations and think tanks ($n = 9$) (see Appendix B in the online material). Given the study's strategic focus, elite interviews (Teixeira and Bridge 2024) were particularly valuable for accessing decision-makers directly involved in shaping or responding to industrial and geoeconomic strategies. Informants were identified through professional networks, targeted outreach, and snowball sampling. We sought diversity in institutional affiliation and position within the value chain to avoid bias. Interviews were conducted in Mandarin, English, or German. About

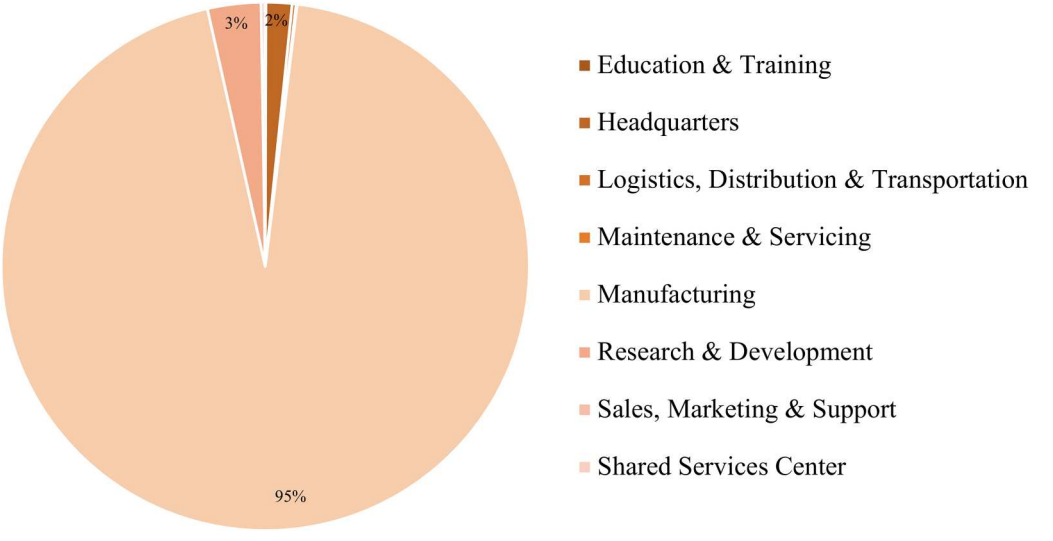


Figure 3. Distribution of Chinese EV greenfield investments in Europe by operational function over the period 2014–23, based on the share of cumulative greenfield investments.
Source: FDI markets; authors' calculation.

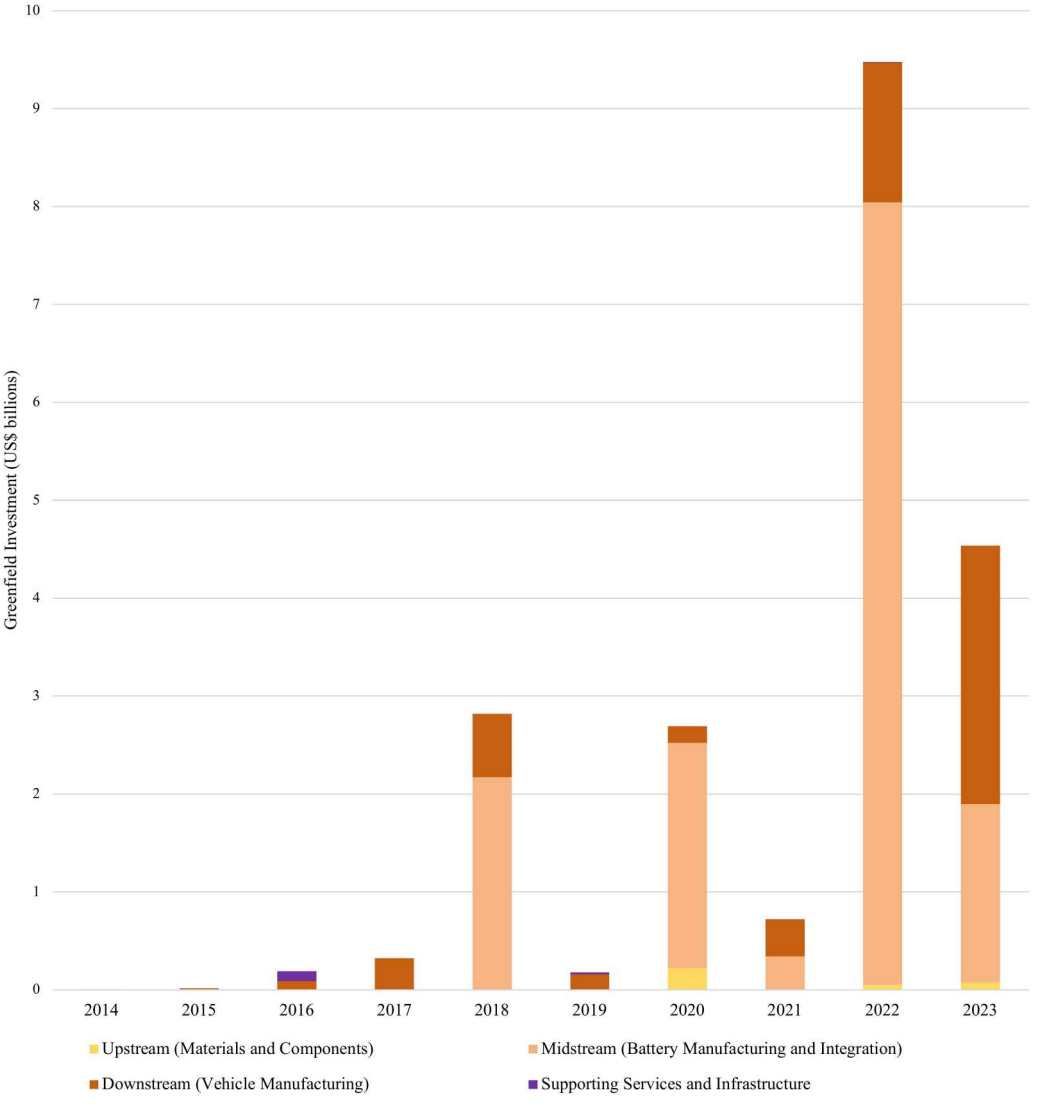


Figure 4. Breakdown of Chinese EV greenfield investment in Europe by value chain segments (billion US dollars).
Source: FDI markets; authors' calculation.

one-third (mainly firm executives and officials) informed the core analysis of firm strategies and state agendas; the remainder provided contextual depth and triangulation. Cited interview data reflect either strategic insights, representative views, or notable divergences. In parallel, we analyzed secondary sources to trace evolving agendas and geoeconomic measures in China and the EU (see Appendix C in the online material), including policy documents, government white papers, media reports, and industry analyses. These materials provided context, cross-verified interview claims, and supported mapping of the policy instruments shaping GPNs. Data analysis proceeded in four steps: (1) examining Chinese and EU agendas and instruments; (2) analyzing their

interplay; (3) exploring how cost-capability, market, and financial dynamics shaped Chinese firms' European strategies; and (4) assessing firms' responses to geoeconomic pressures at home and abroad.

Chinese EV Battery Firms Navigate Between the Agendas of China and the EU

Following the analytical framework proposed earlier, this section empirically analyzes the interplay between states and firms through Chinese EV battery firms' expansion in Europe.

States as Geoeconomically Competing and Strategically Selective Actors: China and the EU in Comparison

Table 1 gives an overview of the agendas and the geoeconomic measures adopted by Chinese government and the European Commission.

The Role of the Chinese Government in Shaping the Emerging EV Battery GPN. China's state project in the EV battery GPN exemplifies a geoeconomically competing logic. The Chinese government has pursued long-term strategies to achieve green technological leadership, strengthen industrial capabilities, and project influence through global value chains. While these goals are not formally articulated as a single, unified state blueprint, they can be understood analytically from China's policy practices and strategic behavior. Our analysis suggests that China seeks to consolidate domestic dominance while extending control over critical nodes of the international battery supply chain. At the same time, China acts in a strategically selective manner, deploying a broad yet coordinated mix of industrial, trade, financial, and foreign policy instruments to advance these aims. Based on observable policy orientations and accumulated evidence, we identify four interrelated agendas that capture the direction of China's state project: (1) establishing green leadership, (2) developing green products as export and growth engines, (3) shaping a global green image, and (4) maintaining industrial leadership while preventing technological leakage. These agendas are realized through a targeted portfolio of geoeconomic measures that, taken together, reinforce China's centrality in the global EV battery value chain. First, the expansion of China-led EV battery GPN has been driven by the government's strategic aim to establish green leadership and achieve technological sovereignty through corner overtaking. In the early development phase (pre-2015), EVs and key components like power batteries and drive motors were prioritized (Liu et al. 2011), and in 2015, EVs were further recognized as a national strategic emerging industry (Gov.cn 2010). Post-2015, to modernize industrial capabilities, EV batteries were designated as a strategic focus. Collectively, industrial policies underscore the Chinese government's ambition to lead in green technology while safeguarding technological sovereignty. Guided by such broader strategic industrial policies, the state also implemented targeted initiatives for the entire EV battery sector (for details, see Gong and Hansen 2023). Moreover, as critical minerals are essential for a globally competitive EV battery industry, the Chinese government has long recognized their strategic importance and prioritized securing supplies to support industrialization (see Economy and Levi 2014). It has pursued a dual strategy, leveraging

Table 1

Geoeconomic Measures in China and the EU in Supporting Their Agendas

Economies	Agendas	(Geo)Economic Measures
China	Establishing green leadership to achieve strategic advancement	**Industrial policy**: the Outline of the National Medium- and Long-term Scientific and Technological Development Plan (MLP); the *Made in China 2025* (MIC25) policy; the 2015 Automotive Power Battery Industry Specification Conditions (the Whitelist)
		Innovation and foreign policy in securing minerals: State Key Laboratories for critical materials established in the 1980s; *Going Global* strategy in the 11th FYP; *Belt and Road Initiative* (BRI); Action Plan for Building a World-Class Salt Lake Industrial Base in Qinghai
	Developing green products as a new export and growth engine	**Industrial policy**: 863 Program (national high-tech R&D program) in 1986; *Mid- to Long-term Development Plan for the Automotive Industry*
		Trade policy: *Opinions on Supporting the Healthy Development of NEV Trade Cooperation*
	Global green image shaping	**Foreign policy**: BRI; BRI International Green Development Coalition (BRIGC)
	Maintaining industrial leadership while preventing technological leakage	**Industrial and innovation policy**: the *Lithium-Ion Battery Industry Specification Conditions (2024 Edition)*
		Export restriction: the MoC approved the revision of the *Catalogue of Technologies Prohibited or Restricted from Export in China*, adding battery cathode material preparation technologies and lithium extraction–related technologies to the list of controlled exports (MoC 2025).
EU	Technological catch-up to address economic security concerns	**Industrial policy**: The *European Green Deal;* the *European Battery Alliance* (EBA); Two *Important Projects of Common European Interest* (IPCEIs); *Green Deal Industrial Plan (including Net Zero Industry Act* and *Critical Raw Materials Act); Innovation Fund (IF24) Battery Calls*
		Innovation policy: BATTERY 2030+
	Advancing decarbonization and green transitions	**Environmental policy**: *European Climate Law*; *Fit for 55 Package; New Battery Regulation*
		Industrial policy: The *European Green Deal*
	Protecting the domestic industry while ensuring a level playing field	**Trade and investment policy:** *FDI Screening Regulation*; the *Foreign Subsidy Regulation (FSR) 2023*; various trade defense tools, including antisubsidy and antidumping measures
		State aid rule adaptation: *Temporary Crisis and Transition Framework* (2019) to support sectors key to accelerate the green transition amid crises

Note: Policies collected here are measures with clear geostrategic objectives, for an articulation of domestic industrial policies on EV batteries in the Chinese policy landscape; see Gong and Hansen (2023).

both domestic resource exploitation and access to foreign markets. Since the early 2000s, Beijing has issued a series of policies to support this agenda. Notably, the *Going Global* strategy, launched in 2000, encouraged Chinese state-owned firms, among other objectives, to invest in resource-rich countries, while the Belt and Road Initiative (BRI), announced in 2015, aimed to establish logistical corridors for mineral trade and support upstream investments abroad. As global competition for resources has intensified, the central government has introduced new measures, including the *Raw*

Materials Industry Development Plan under the Fourteenth Five-Year Plan, to ensure stable supply for both domestic needs and the global EV market. These policies reflect a strategically selective orchestration of domestic capacity-building and overseas resource acquisition to secure the material foundations of industrial leadership.

Second, the state has encouraged firms to expand outward and build cross-border production capabilities. This has enabled Chinese firms to strategically couple with host economies, especially in Europe. Specifically, Chinese brands were expected to increase their overseas influence by 2020, with several Chinese automotive firms projected to be among the global top ten by 2025. More recently, policies further promote EV trade cooperation such as strengthening financial support and encouraging banking institutions to better meet the financial needs of EV firms (MoC 2023). These policies reflect an intentional strategy to extend the territorial reach of the Chinese EV battery production network, secure downstream markets, and mitigate trade frictions, while retaining control over design, technology, and upstream inputs. This transnational expansion is often supported through coordinated state–finance mechanisms and policy facilitation, reinforcing the state's role in shaping the extraterritorial architecture of the GPN.

Thirdly, the Chinese government has also worked to legitimize its central role in the green technology GPN through building up international communication capacity and formulating diplomatic policies. Since 2015, China has aimed to foster a green image through international initiatives like the BRI. Recognizing the environmental impact of large-scale development, China further launched the BRI International Green Development Coalition (BRIGC), which aims to enhance China's green credentials globally (UN Environment Programme 2019). Increasingly, China's export of lithium-ion batteries, EVs, and photovoltaic products, dubbed the new trio, is presented by Chinese leaders as aligning with the global sustainability objectives, allowing China to present its GPN leadership as mutually beneficial and compatible with frameworks like the EU Green Deal, thereby reinforcing the global embeddedness of Chinese firms and technology standards.

Finally, as China's leading firms climb to higher-value segments of the GPN, the state has shifted its focus toward preserving national advantage and preventing technological outflows. To maintain leadership, the MIIT recently issued the *Lithium-ion Battery Industry Specification Conditions 2024 Edition* (MIIT 2024). It requires firms to establish independent R&D institutions at or above the provincial level, to apply for technological invention patents for their main products, and to ensure that annual expenses for R&D and process improvement are no less than 3 percent of their main business income (MIIT 2024). Incentivized by this, China plans to invest more than six billion yuan ($830 million) in a government-led project to develop the next-generational solid-state batteries in collaboration with six firms, including Contemporary Amperex Technology Co., Ltd. (CATL) and BYD Company Limited (BYD) (Yan et al. 2024). The Chinese state is also concerned about the potential for technological leakage as domestic firms expand globally. In mid-2025, the MoC approved the revision of *the Catalogue of Technologies Prohibited or Restricted from Export in China*, adding battery cathode material preparation technologies and lithium extraction–related technologies to the list of controlled exports (MoC 2025). With the adoption of these controls, China has effectively closed the window on unfettered export of critical battery technology, reinforcing its

strategic grip on the EV battery GPN. Such measures illustrate how China's state project combines competitive geoeconomic ambition with selective governance, using a wide repertoire of tools, to advance its strategic objectives.

The EU's Concern on Technology Sovereignty and Economic Security in the EV Battery GPN. The EU's approach to the EV battery industry likewise reflects a geoeconomically competing stance, though rooted in its position as a late entrant. Confronted with growing dependence on Asian firms across the value chain, the EU's agendas can be summarized in three categories based on the analysis of its policy documents: (1) technological catch-up to address economic security concerns, (2) advancing decarbonization and green transitions; and (3) protecting the domestic industry while ensuring a level playing field. It has mobilized a comprehensive policy framework to reach such agendas.

The *European Green Deal*, initiated in 2019, underscores the EU's commitment to decarbonization and strategic autonomy. It aims to make Europe the first climate-neutral continent by 2050 (European Commission 2019a). To support this ambitious climate goal, the *European Climate Law* was passed in 2021, enshrining the 2050 climate neutrality target into EU legislation (European Parliament and Council 2021). Furthermore, the *Fit for 55 Package* was also announced in 2021, including proposals to align EU policies with the 2030 climate target of a 55 percent reduction in greenhouse gas emissions (European Commission 2021). While presented primarily as environmental policies, these initiatives also function as geoeconomic instruments, aligning decarbonization with industrial competitiveness (Meckling 2025). Industry-specific measures, such as the *New Battery Regulation*, introduce sustainability criteria, including carbon footprint reporting and battery passports, aligning not just with environmental aims but also creating entry barriers for global battery producers (European Parliament and Council 2023a). The Battery Passport, in particular, has emerged as a crucial governance tool in the GPN of battery production, influencing standards and access in the focal GPN.

From a GPN perspective, the EU's battery agenda reflects an attempt to reconfigure value chains to favor local production and capture more value domestically. Battery technology, as a strategic node in the broader EV production network, plays a central role in Europe's low-carbon transition. Yet, EU firms remain dependent on upstream and midstream segments controlled by Asian producers (IEA 2024). In response, the EU aims to insert itself more effectively into these networks and to anchor more stages of the value chain territorially, through policies designed to stimulate R&D, scale production, and secure critical inputs.

Launched in 2017, the *European Battery Alliance* (EBA) aims to consolidate Europe's position in battery technology, reducing reliance on Asian suppliers. The EU has also prioritized the battery supply chain as a matter of national security, funding initiatives like the *BATTERY 2030 +* research program, and two *Important Projects of Common European Interest* (IPCEIs) were granted with €3.2 billion and €2.9 billion public funding respectively (European Commission 2020, 2019b), to support innovation across the value chain as well as the upscaling of production, which was further bolstered by the *Green Deal Industrial Plan* (European Commission 2023a).

Despite this push, the EV battery production networks in Europe remain highly interconnected with Chinese actors. These investments illustrate that Europe's battery production network is not an isolated endeavor—it is embedded in global circuits of capital,

knowledge, and components, often steered by lead firms headquartered in China. While these Chinese firms bring production closer to EU markets and consumers, they also raise concerns about dependency and control, leading to more geoeconomic scrutiny and regulatory action such as through the *FDI screening regulation* (EU Regulation 2019/452).[2]

To further assert regulatory control over these production networks, the EU has activated a number of geoeconomic instruments with GPN effects. The *Foreign Subsidy Regulation* (FSR) (European Commission 2023b) targets market distortions caused by foreign state-backed firms, affecting bidding processes in battery projects involving Chinese entities. Likewise, the *State Aid Temporary Crisis and Transition Framework* allowed Germany to approve a €902 million subsidy to Northvolt, a Swedish battery maker, to build a plant in Heide. This selective support for European firms represents an effort to create lead firm capabilities within the EU and foster more regional value capture.

The GPN logic is further reinforced in the Net Zero Industry Act (European Parliament and Council 2023b) and Critical Raw Materials Act (European Parliament and Council 2024), operationalizing a selective reshoring strategy, linking security of supply to the localization of value creation. Both policies aim to secure upstream control and reduce supply chain risk through onshoring, recycling, and diversification. Moreover, in December 2024, the Commission launched the Innovation Fund Battery Calls, allocating €1 billion for battery cell manufacturing projects, with an explicit objective to de-risk supply chain dependence on China, especially for anode and cathode materials (European Commission 2024). The new *security of supply and countering dependency* award criterion ensures that these projects are evaluated not just on innovation but on how they reshape power and dependencies within the battery GPN. In sum, the EU's geoeconomic measures are not only strategic but constitutive of new GPN architectures. The outcome is not a decoupled network, but rather a reconfigured one, shaped by overlapping industrial, environmental, and security goals.

Geoeconomic Chain Reaction. While the agendas of China and the EU were primarily developed in line with their own strategic interests, there are nonetheless important instances of interstate dynamics within the China–Europe EV battery production networks. Thanks to over a decade of generous subsidies and industrial policy instruments, Chinese firms have secured competitive advantages across almost all segments of the battery GPN, from material processing and cell production to battery pack assembly (Gong and Andersen 2024). This growing dominance has placed significant competitive pressure on European firms and policy makers, compelling a strategic reassessment of the EU's position in the EV battery GPN. In response, the EU has, since 2017, introduced a series of industrial and innovation policies aimed at reshaping the territorial embeddedness of the battery GPN within Europe. These initiatives, including the EBA, the IPCEI framework, and more recently the IF24 Battery Call, aim not only to foster local champions but also to build out a European production network capable of capturing greater value and exerting more control over critical supply chains. Importantly, these policies have not only supported local firms but have also pressured leading

[2] Regulation (EU) 2019/452 of the European Parliament and of the Council —of March 19, 2019—establishing a framework for the screening of FDIs into the European Union.

Chinese players to reconfigure their GPN strategies by establishing production facilities and R&D centers within the EU to maintain market access and political legitimacy.

A second and more overt instance of a geoeconomic chain reaction emerged through recent trade defense measures. Supported by the EU FSR and broader trade instruments, the European Commission initiated an antisubsidy investigation in September 2023 targeting Chinese EV imports. This move, justified as a response to alleged market distortions caused by non-EU state support, aimed to protect the competitiveness of the European automotive sector. The resulting tariffs, implemented on October 30, 2024, exemplify how state-led interventions reshape GPN governance, often forcing lead firms to reassess their regional embeddedness and cross-border strategies. In retaliation, China launched its antidumping investigations targeting European goods, such as brandy and luxury products, and imposed tariffs of up to 39 percent on brandy imports. These measures underscore how state actions reverberate through GPNs, influencing not only firm behavior but also the spatial configuration of production, investment, and trade relations.

Firms as Geoeconomic Actors: Chinese EV Battery Firms' Strategies in Balancing Geopolitical Concerns and Business Interests

Intensified geoeconomic competition has significantly influenced the operations of Chinese lead firms. While these firms continue to operate within the competitive dynamics that Coe and Yeung (2015) identify, the relative importance and functioning of these factors appear to diverge from the patterns observed in more established GPNs of the neoliberal globalization era. Additionally, geopolitical risks, which are not accounted for in Coe and Yeung's (2015) original framework, now play a pivotal role in shaping the strategies of Chinese lead firms in Europe. The remainder of this section explores how these forces manifest in the emerging EV battery GPN.

The Recalibration of Competitive Dynamic Forces by Chinese EV Battery Firms in Europe. Among the three competitive dynamic forces shaping firms' internationalization strategies, the expansion of Chinese firms into the EU is primarily driven by the market imperative. With EVs comprising over 40.9 percent of new car sales in China in 2024 (Zhang 2025), intensifying competition has sharply eroded profit margins, compelling firms to seek international opportunities for expansion. The sentiment *go overseas or be out of the game* strongly resonates within the industrial practitioners, underscoring the urgency of establishing a global presence. Despite challenges, the European market presents a compelling opportunity, being more open than other major markets: "While EU consumers demand higher product quality and stricter environmental standards, these expectations also promise higher profit margins. . . . For firms like ours, accustomed to intense price competition in the domestic market, entering markets that prioritize premium products presents a promising pathway to long-term growth" (Interview, Industry Representative No. 16, September 13, 2023, Shenzhen). This trend is evident in the performance of leading Chinese battery firms that experience substantial growth in overseas markets. CATL, for instance, reported overseas revenue of RMB 67.17 billion in the first half of 2023, representing a year-on-year increase of 195.15 percent. This growth far outpaces the 46.10 percent increase in its domestic market, with overseas income accounting for 35.49 percent of its total revenue (CATL 2023).

In terms of cost-capability ratios, the expansion of Chinese lead firms into Europe deviates from the production flow patterns that Coe and Yeung (2015) hypothesize, which favors lower cost-capability ratios to maintain competitiveness. Expanding into Europe tends to increase these ratios for Chinese lead firms, given the high costs and relatively low capabilities in battery manufacturing within the region. In 2024, average battery pack costs in Europe were 48 percent higher than in China, where the cost was $94/kWh (BloombergNEF 2024). At the same time, Europe's innovation performance in this field lags behind: While China's global share of lithium-ion battery patents grew from 2.3 percent in 2010 to 15.8 percent in 2023, Europe's share decreased from 12.6 percent to 5.3 percent during the same period (Appendix D in the online material). Despite these disadvantages, Chinese firms are still expanding into Europe. This decision is largely driven by the market expansion imperative and the increasing adoption of geoeconomic measures by the EU and its member states to attract lead firms to settle down in Europe. "If there were no trade wars or tariff disputes, we would prefer to produce domestically and export to overseas markets. Firstly, we have a complete industrial chain, skilled workers, and access to raw materials in China. Secondly, domestic production efficiency is generally high. . . . However, in the face of geopolitical risks, these production advantages are no longer the most critical factors" (Interview, Industrial Representative No. 12, December 18, 2023, Hefei). Additionally, securing alternative markets has become crucial for sustaining business operations in an intensely competitive global market. As a top manager from an electrolytes company explained, "This is our first time venturing into the European market. If the international situation were relatively stable, we might have spent more time on strategic assessment. However, with the EU tightening its trade and investment policies toward Chinese firms, we must act quickly to establish a presence in Europe to mitigate the impact of geopolitical factors on our long-term business plans" (Interview Industrial Representative No. 14, May 25, 2024, Hannover). Such limited preparedness of Chinese emerging leaders to operate within culturally and politically distinct contexts has become evident. As Langguth (2024) shows, the experiences of Chinese EV battery firms in Germany reveal how frictions with local institutions, regulatory frameworks, and community expectations can complicate project implementation. These conflicts highlight that expanding abroad under geoeconomic pressure requires not only cost-capability considerations but also institutional sensitivity and cross-cultural adaptability—competencies that many Chinese firms are still developing.

Finally, financial discipline profoundly influences the strategies of Chinese EV battery firms, shaping their pace of growth, modes of market entry, and funding mechanisms. Two aspects illustrate this dynamic. First, the expansion of Chinese EV battery firms in Europe requires striking a delicate balance between rapid growth and the substantial capital investments needed for overseas operations. For example, Svolt Energy canceled plans for two battery factories in Germany due to financial constraints (Xie 2024). This decision underscores the significant and sometimes unexpected costs of greenfield investments in Europe, such as land acquisition, labor, and compliance with stringent environmental and regulatory requirements (Langguth 2024). Learning from this experience, Chinese firms adopt a diversified approach to market entry including small-scale R&D centers or pilot production facilities (e.g., Gotion High-tech Co., Ltd. [Gotion]), joint ventures (Gotion with Volkswagen, Leapmotor with Stellantis),

greenfield investments (CATL, BYD, Gotion) to diversify the financial risks (Gong and Andersen 2024).

Another key aspect of financial discipline for Chinese EV battery firms is their ability to strategically leverage domestic and EU financial opportunities. "Chinese firms can choose to establish production facilities in EU countries with the most favorable incentives while exporting components or raw materials from China to take advantage of export tax returns" (Interview, Industrial Representative No. 21, January 18, 2024, Berlin). On the domestic front, Chinese firms benefit from export tax rebates and subsidies. These incentives support overseas production and export activities, helping firms offset the high costs of expanding abroad. Moreover, Chinese EV battery makers also benefit from state-backed credit and equity financing that is cheaper than regular offers on the financial markets, enabling them to scale up production capacity over and above their commercial means (Sebastian 2023). In the EU, Chinese firms also strategically leverage state support and incentives offered by host countries. Many EU member states, particularly those with ambitions to become battery manufacturing hubs, provide substantial subsidies, tax incentives, and grants to attract foreign investment.

Firms' Geoeconomic Strategies. Chinese EV battery firms are increasingly adopting a diverse range of strategies to navigate both the domestic and EU policy landscapes. Table 2 gives an overview of such strategies.

Active Lobbying and Selective Noncompliance. In China, some firms have responded cautiously to nonbinding state recommendations. A recent example is guidance from the MoC advising manufacturers to keep key component production within China and limit overseas outsourcing to final assembly in order to prevent technological leakage (Lew 2024). While acknowledging the rationale, many firms emphasized the need for global operations in an overcrowded domestic market. As one executive from CATL explained, "International collaborations do not inherently threaten Chinese technological sovereignty" (Interview Industrial Representative No. 4, December 3, 2021, Ningde). Firms, such as CATL, BYD, Nio, and Gotion, have therefore prioritized building innovation and production bases overseas, selectively implementing state guidance.

In the EU context, Chinese firms and industry associations have lobbied against proposed trade restrictions. After the EU's 2025 decision to impose five-year tariffs on China-imported EVs, the China Chamber of Commerce to the EU (CCCEU) published a survey showing that 82 percent of Chinese firms reported reduced investment confidence (CCCEU 2024). The report sought to influence policy debate by highlighting how the tariffs undermine Europe's own green-transition objectives and overstate the so-called China challenge. The China Association of Automobile Manufacturers (CAAM) and the MoC similarly expressed concerns that tariffs would harm investment and local employment in Europe (*Global Times* 2024). Chinese executives also participated directly in this discursive process, supporting the lobbying and negotiation of the Chinese authorities and associations (CCCEU 2024). These actions exemplify active lobbying and selective noncompliance: strategies grounded in the production and mobilization of knowledge to influence the design and direction of pending or nonmandatory policies. While firms cannot ignore binding rules once enacted, they employ research, data, and negotiation to shape how such measures evolve and are ultimately implemented.

Table 2

Chinese Firms' Geoeconomic Strategies at Home and Abroad

Strategies	Applicable Type of Geoeconomic Measure	Core Mechanism and Rationale	Illustrative Examples
Active lobbying and selective noncompliance	Nonbinding, proposed or recommendatory measures issued by home or host states	Firms seek to influence, delay, or selectively implement nonmandatory guidance. This includes rhetorical opposition, lobbying, or procedural negotiation.	***Domestic political arena***: –Limited adherence to MoC guidance on producing key components domestically –Public appeals for flexible implementation by executives ***International corporate statecraft***: –Participation in industry associations' consultations and reports, urging dialogue on tariffs and lobbying against EU antisubsidy measures
Compliance and adaptation	Legally binding regulations and trade or investment rules that condition market access	Firms comply with binding rules but adapt operations and ownership structures to mitigate costs and preserve competitiveness.	***Domestic political arena***: –CATL restructuring overseas financing to comply with China's outward foreign direct investment (OFDI) and capital-control regime –Public listing in Switzerland and Hong Kong to circumvent strict OFDI regulations –Adjustment to China's strategic technology export-control rules by license applications and localization of R&D ***International corporate statecraft***: –Compliance with EU environmental and labor standards while adjusting production processes –Shift toward exporting hybrid or plug-in-hybrid (PHEV) vehicles to Europe to avoid EU tariffs targeting only BEVs –Locating in Hungary/Poland or forming EU joint ventures to satisfy FDI-screening requirements
Strategic alignment	Policies whose objectives coincide with firms' commercial interests	Firms proactively align with domestic or foreign state initiatives to gain financing, legitimacy, or market advantage, using alignment as a business strategy.	***Domestic political arena***: –Participation in the BRI and BRI-linked EV projects –Compliance with and benefit from the *Lithium-ion Battery Industry Specification Conditions (2024 Edition)* through increased R&D investment ***International corporate statecraft***: –Framing CATL's European investments as contributing to the EU Green Deal and *Fit for 55* agenda

(*Continued*)

Compliance and Adaptation. When measures are binding, Chinese firms comply but adapt financing, production, and governance structures to sustain competitiveness. In the domestic political arena, China's mixed stance on outward foreign direct investment

Table 2 *Continued.*

Strategies	Applicable Type of Geoeconomic Measure	Core Mechanism and Rationale	Illustrative Examples
Advocacy and policy shaping	Future regulatory frameworks or industrial policies that have not yet been formulated, finalized, or enacted	Firms collectively or individually influence future policy design to secure supportive conditions for internationalization.	*Domestic political arena:* –Calls by JAC Motors and others for public-service platforms supporting overseas expansion *International corporate statecraft:* –Investment siting aligned with EU regional-development and cohesion-policy incentives to secure subsidies and local legitimacy

(OFDI) *Lithium-ion Battery Industry Specification Conditions* creates simultaneous incentives and constraints. The state encourages global expansion through credits, subsidies, and tax benefits, while ministries maintain tight screening to prevent excessive capital outflow and technology loss. CATL's decision to seek direct investment from Norway's Sovereign Wealth Fund for its European projects exemplifies adaptation to these restrictions (Wiggins et al. 2024). Likewise, many Chinese EV battery firms have turned to listings in Switzerland and Hong Kong to raise capital abroad while keeping domestic funds within China. "Listing on overseas Stock Exchange aligns with the central government's encouragement to 'go global' … it also enables Chinese capital to remain within China" (Interview Industrial Representative No. 10, August 21, 2023, online). Firms have also adjusted to tightening export-control rules. As a response to the MoC's decision to formally add battery-cathode material preparation and lithium-extraction technologies to the *Catalogue of Technologies Prohibited or Restricted from Export*, firms have complied by applying for licenses and localizing R&D to continue international collaboration under new restrictions.

In Europe, compliance and adaptation involves meeting the EU's stringent environmental and labor standards. "We are fully committed to the European market. However, environmental regulations in the EU can be overwhelming, given the limited timeframe we have to adapt to them," noted one manager (Interview Industrial Representative No. 3, December 15, 2021, Shenzhen). Firms have invested in cleaner production and reconfigured supply chains accordingly. Moreover, as a deliberate response to EU tariffs that only apply to pure battery-electric vehicles, many Chinese original equipment manufacturers (OEMs) have increased exports of hybrid or plug-in-hybrid (PHEV) models, thereby sidestepping tariff regimes that exclude those powertrains. For instance, Chinese automakers exported 3,269 PHEVs to the EU in March 2025, up from nearly zero before the tariff announcement (Parodi 2025). To further mitigate regulatory risks, Chinese investors have selected production sites in Hungary, Spain, and Poland or established joint ventures with European OEMs to satisfy the FDI screening regulation and ease political scrutiny (Sebastian, Goujon, and Meyer 2024). Together, these examples illustrate how Chinese firms combine formal compliance with strategic flexibility, reshaping their operations within the constraints of an increasingly politicized regulatory environment.

Strategic Alignment with Geoeconomic Measures. When commercial and political goals coincide, firms engage in strategic alignment with state agendas. Many firms actively support China's industrial and geopolitical ambitions by participating in BRI projects in Southeast Asia, Africa, and Europe. They also adhere to the *Lithium-ion Battery Industry Specification Conditions (2024 Edition)*, which encourage firms to invest more than 3 percent of annual income in advanced battery R&D. By doing so, firms access state funding and maintain leadership in global battery innovation. Moreover, Chinese EV battery producers contribute to the consolidation of domestic supply chains, a key component of the country's industrial resilience strategy. This involves strengthening partnerships with domestic suppliers of critical raw materials, and investing in recycling technologies, to mitigate risks associated with import dependencies and global price fluctuations (Sebastian 2023). They also reinforce the government's push for regional rebalancing by establishing facilities in inland provinces to promote green industrialization and green growth in China's underdeveloped regions.

In the EU, Chinese firms reinforce EU objectives by promoting their products and investments as essential to Europe's green transition. For example, several EV battery firms have strategically positioned their investments in the EU as beneficial to the bloc's climate goals. This reinforcement strategy helps the firm navigate the EU's political landscape, creating a narrative that positions Chinese firms as essential partners in achieving the EU's environmental ambitions. It also helps to offset potential resistance or restrictions tied to foreign investment. Chinese firms also invest in greenfield projects and R&D centers to align with Europe's emphasis on innovation and sustainability. By doing so, they support the EU's broader green transition goals, which often come with incentives such as subsidies, grants, and tax breaks for sustainable industrial development.

Advocating and Policy Shaping. Chinese EV battery firms also advocate for greater support both domestically and in the EU, though the types of assistance they can seek are more diverse at home than abroad. In the domestic political arena, Chinese firms have pleaded for various support to reduce their cost-capability ratios and thus boost their global competitiveness. As Chinese EV battery producers are relatively new to international markets, most of them lack overseas sales, supply, and service networks. Moreover, the increasing politicization of trade has added more difficulties to their global operation (Sebastian 2021). Consequently, they have called for state support to aid their global operations. For example, Li Ming, General Manager of JAC Motors, suggested that auto- and battery-makers work together with the state to build a comprehensive public service platform to provide full-spectrum assistance to Chinese firms expanding internationally (Zhang 2024). Other firms have proposed several additional measures at the regulatory and supply chain levels (e.g., supporting private firms through financial and tax incentives). More recently, Chinese EV battery firms also lobbied the government to engage diplomatically with the EU to address tariffs and other export restrictions.

In the EU, Chinese firms selectively plead with specific EU member states that maintain more favorable attitudes toward Chinese investment to provide more accommodating policies. A notable example of Chinese firms advocating for support within the EU is their engagement with regional development programs in structurally weak areas. By targeting investments in regions that are eligible for European Structural and Investment Funds (EU's Cohesion Policy), and by aligning their projects with local development

goals, they benefit from incentives aimed at boosting employment and economic growth in these regions.

Discussion and Conclusion

Through an in-depth analysis of how Chinese EV battery firms navigate the agendas of China and the EU amid intensified geoeconomic competition, this article develops a conceptual–analytical framework that captures the complexities of emerging strategic GPNs. The framework serves as a heuristic tool for uncovering the political and strategic forces reshaping global production. It reflects a broader transformation from firm-led globalization to hybrid state–firm driven dynamics in strategic emerging GPNs, where production is co-constituted through intertwined political, strategic, and economic logics. This article advances the GPN framework's relevance for understanding strategic sectors led by nonconventional firms in three key ways.

First, it offers a nuanced analysis of the state as a geoeconomically competing and strategically selective actor in strategic emerging GPNs (Hess and Horner 2024; Coe et al. 2025; Li and Gong 2025). By integrating neo-Gramscian theory, strategic-relational approach, and insights from geoeconomics, we examine both why states pursue particular agendas via strategic articulations of economic, political, security, and symbolic objectives in the global arena, and how they do so via the strategic deployment of geoeconomic instruments. Geoeconomics enriches this analysis by highlighting the growing use of economic tools to achieve geopolitical goals, particularly in sectors marked by technological rivalry and strategic interdependence. We highlight contrasting state projects: China's pursuit of industrial leadership, green exports, and image making as a first mover versus the EU's fragmented quest for technological catch-up, economic sovereignty, and sustainability, partly shaped by civil-society pressures. We conceptualize their interaction as *geoeconomic chain reactions*—mutually adaptive state responses enacted through industrial, innovation, trade, and environmental policies. This approach clarifies the strategic motivations and instruments of state actors, addressing calls for deeper engagement with critical state theory in GPN research (Coe et al. 2025).

Second, our study advances understanding of firm behavior under geoeconomic competition (Choer Moraes and Wigell 2020; Li and Gong 2025; Lim 2025). Whereas prior GPN research centered on advanced-economy lead firms under neoliberal globalization (Horner and Nadvi 2018), EMNEs now operate amid geopolitical rivalry and selective integration (Li and Gong 2025). Their strategies must reconcile economic goals with geopolitical imperatives, often relying on home-state support that reinforces tighter state–firm linkages (Guo et al. 2023). Building on the notion of firms as geoeconomic actors (Ferguson, Moraes, and Wigell 2025; Lim 2025), we identify four strategic responses, that is, active lobbying and selective noncompliance, compliance and adaptation, strategic alignment, and advocacy, that unfold across home- and host-country political arenas. This detailed analysis offers a more differentiated account of firm agency under geopolitics than in GPN 2.0.

Third, building on the competitive dynamic forces that Coe and Yeung (2015) identify, our analysis demonstrates that geoeconomic factors significantly shape how these forces operate within a strategic emerging GPN. For Chinese EV battery firms, their expansion into the EU is primarily driven by the market imperative. This market

imperative has prompted Chinese MNEs to adopt strategies with higher cost-capability ratios, a notable departure from the lower cost-capability strategies typical of the neoliberal globalization era. Additionally, the pursuit of financial discipline by Chinese EV battery firms bears geopolitical imprints. Stringent regulations on OFDI imposed by the Chinese state, coupled with high standards often required by European stakeholders to leverage EU incentives, further complicate their financial strategies. These dynamics highlight the need to reinterpret the assumptions underpinning the three dynamic forces, which were originally developed in the context of neoliberal globalization, to account for the contemporary geopolitical and geoeconomic landscape.

By blending the analytical strengths of GPN and geoeconomics, our framework not only captures the nuances of Chinese MNEs operating under global pressures but also potentially serves as a heuristic to analyze other cases where emerging-market firms are reshaping production networks in response to, and as agents of, geoeconomic strategies. Future research could deepen this analysis by comparing Chinese lead firms' strategies across the EU, the US, and other regions, or by examining how latecomer governments respond to China's industrial leadership (Bridge and Faigen 2022). Comparative studies across sectors and varieties of capitalism would clarify how institutional settings shape firms' geoeconomic agency and degrees of autonomy (Henderson and Hooper 2021). Finally, future work should explore the multiscalarity of geoeconomic governance, or how state selectivity and competition unfold not only at national and supranational levels but also through provincial, city–regional, local configurations, thereby revealing the differentiated geographies of geoeconomic power.

References

Alami, I. 2024. Foreign investment screening mechanisms and emergent geographies of (post)globalization. *Dialogues in Human Geography*. doi: 10.1177/20438206241278733.

Alami, I., and Dixon, A. D. 2024. *The spectre of state capitalism*. Oxford University Press.

Aoyama, Y., Song, E., and Wang, S. 2024. Geopolitics and geospatial strategies: The rise of regulatory supply chain controls for semiconductor GPN in Japan, South Korea and Taiwan. *ZFW—Advances in Economic Geography* 68 (3–4): 167–81. doi:10.1515/zfw-2024-0046.

Binz, C., Gosens, J., Yap, X. S., and Yu, Z. 2020. Catch-up dynamics in early industry life-cycle stages—A typology and comparative case studies in four clean-tech industries. *Industrial and Corporate Change* 29 (5): 1257–75. doi:10.1093/icc/dtaa020.

Blackwill, R. D., and Harris, J. M. 2016. *War by other means: Geoeconomics and statecraft*. Harvard University Press.

Blažek, J., and Lypianin, A. 2024. Geopolitical decoupling and global production networks: The case of Ukrainian industries after the 2014 Crimean annexation. *Journal of Economic Geography* 24 (1): 23–40. doi:10.1093/jeg/lbad026.

BloombergNEF. 2024. Lithium-ion battery pack prices see largest drop since 2017, falling to $115 per Kilowatt-Hour. December 10. https://about.bnef.com/insights/commodities/lithium-ion-battery-pack-prices-see-largest-drop-since-2017-falling-to-115-per-kilowatt-hour-bloombergnef.

Bridge, G., and Faigen, E. 2022. Towards the lithium-ion battery production network: Thinking beyond mineral supply chains. *Energy Research & Social Science* 89: art. 102659. doi:10.1016/j.erss.2022.102659.
CATL. 2023. CATL 2023 semiannual report. https://www.catl.com/uploads/1/file/public/202307/20230726085836_osr3qig4p7.pdf.
China Chamber of Commerce to the EU. 2024. Greening Europe: Report on the development of Chinese NEV manufacturers in Europe. https://weshare.cnfic.com.cn/resource/activity/activity40/20240712/Greening%20Europe%20Report%20on%20the%20Development%20of%20Chinese%20NEV%20Manufacturers%20in%20Europe.pdf.
Choer Moraes, H., and Wigell, M. 2020. The emergence of strategic capitalism: Geoeconomics, corporate statecraft and the repurposing of the global economy. Working Paper No. 117. Finnish Institute of International Affairs.
Coe, N. M., and Gibson, C. 2023. Progress in economic geography? Decarbonising global production networks (GPNs). *Progress in Economic Geography* 1 (1): art. 100002. doi:10.1016/j.peg.2023.100002.
Coe, N. M., Sinclair, L., Gibson, C., and Warren, A. 2025. Resourcing GPNs: Multi-scalar state derisking of energy transition minerals at a time of polycrisis. *Journal of Economic Geography*. doi: 10.1093/jeg/lbaf020.
Coe, N. M., and Yeung, H. W-c. 2015. *Global production networks: Theorizing economic development in an interconnected world*. Oxford University Press.
Coe, N. M., and Yeung, H. W-c. 2019. Global production networks: Mapping recent conceptual developments. *Journal of Economic Geography* 19 (4): 775801. doi:10.1093/jeg/lbz018.
Cox, R. W. 1983. Gramsci, hegemony and international relations: An essay in method. *Millennium* 12 (2): 162–75. doi:10.1177/03058298830120020701.
Curran, L. 2015. The impact of trade policy on global production networks: The solar panel case. *Review of International Political Economy* 22 (5): 1025–54. doi:10.1080/09692290.2015.1014927.
Economy, E., and Levi, M. 2014. *By all means necessary: How China's resource quest is changing the world*. Oxford University Press.
Edler, J., Blind, K., Kroll, H., and Schubert, T. 2023. Technology sovereignty as an emerging frame for innovation policy: Defining rationales, ends and means. *Research Policy* 52 (6): art. 104765. doi:10.1016/j.respol.2023.104765.
European Commission. 2019a. European Green Deal. https://ec.europa.eu/info/strategy/priorities-2019-2024/european-green-deal_en.
European Commission. 2019b. Important Projects of Common European Interest (IPCEIs). https://www.ipcei-batteries.eu/about-ipcei.
European Commission. 2020. BATTERY 2030+ Research Program. https://battery2030.eu/.
European Commission. 2021. Fit for 55 Package. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52021DC0550.
European Commission. 2023a. Green Deal Industrial Plan. https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal/green-deal-industrial-plan_en.
European Commission. 2023b. Foreign Subsidies Regulation. https://competition-policy.ec.europa.eu/foreign-subsidies-regulation_en.
European Commission. 2024. Innovation Fund 2024 Call and Battery Calls. https://cinea.ec.europa.eu/funding-opportunities/calls-proposals/innovation-fund-2024-call-and-battery-calls_en.
European Parliament and Council. 2021. European Climate Law. https://eur-lex.europa.eu/eli/reg/2021/1119/oj.
European Parliament and Council. 2023a. New Battery Regulation. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32023R1542.
European Parliament and Council. 2023b. Net Zero Industry Act. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52023PC0161.
European Parliament and Council. 2024. Critical Raw Materials Act. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32024R1252&qid=1720020986785.

Farrell, H., and Newman, A. L. 2019. Weaponized interdependence: How global economic networks shape state coercion. *International Security* 44 (1): 42–79. doi:10.1162/isec_a_00351.

Ferguson, V. A., Moraes, H. C., and Wigell, M. 2025. States, firms, and geoeconomic agency: The role of private business in economic statecraft and security. *Law & Geoeconomics* 1 (1): 5–23. doi:10.1163/29505720-20250001.

Glassman, J. 2011. The geo-political economy of global production networks. *Geography Compass* 5 (4): 154–64. doi:10.1111/j.1749-8198.2011.00416.x.

Glassman, J. 2025. Critical geoeconomics, critique of geoeconomics or something else? *Environment and Planning A: Economy and Space* 57 (1): 108–12. doi:10.1177/0308518X241270015.

Global Times. 2024. High tariffs on Chinese EVs erode confidence in Chinese investments and operations in Europe, says Chinese automakers association. August 21. https://www.globaltimes.cn/page/202408/1318416.shtml.

Gong, H., and Andersen, A. D. 2024. The role of material resources for rapid technology diffusion in net-zero transitions: Insights from EV lithium-ion battery technological innovation system in China. *Technological Forecasting and Social Change* 200: art. 123141. doi:10.1016/j.techfore.2023.123141.

Gong, H., Hassink, R., Foster, C., Hess, M., and Garretsen, H. 2022. Globalisation in reverse? Reconfiguring the geographies of value chains and production networks. *Cambridge Journal of Regions, Economy and Society* 15 (2): 165–81. doi:10.1093/cjres/rsac012.

Gong, H., and Hansen, T. 2023. The rise of China's new energy vehicle lithium-ion battery industry: The coevolution of battery technological innovation systems and policies. *Environmental Innovation and Societal Transitions* 46: art. 100689. doi:10.1016/j.eist.2022.100689.

Gov.cn. 2010. Interpretation on "The decision of the State Council on accelerating the cultivation and development of strategic emerging industries." http://www.gov.cn/gzdt/2010-10/21/content_1727316.htm.

Guo, H., Chen, Q., Dong, M., Liu, S., Wang, D., and Chen, L. 2024. Risks and responses of China's new energy vehicle industry under cross-border regulations. *King & Wood Mallesions*, May 6. https://www.kwm.com/cn/zh/insights/latest-thinking/cross-border-supervision-risks-and-countermeasures-for-chinese-new-energy-vehicle-companies-part-a.html.

Guo, Y., Poon, J., Wang, C., and Yang, Y. 2023. China's national oil companies going global and coming home: State–firm nexus, global financial networks, and geopolitical risks. *Geoforum* 146: art. 103880. doi:10.1016/j.geoforum.2023.103880.

Henderson, J., Dicken, P., Hess, M., Coe, N., and Yeung, H. W.-c. 2002. Global production networks and the analysis of economic development. *Review of International Political Economy* 9 (3): 436–64. doi:10.1080/09692290210150842.

Henderson, J., and Hooper, M. 2021. China and European innovation: Corporate takeovers and their consequences. *Development and Change* 52 (5): 1090–121. doi:10.1111/dech.12684.

Herranz-Surrallés, A., Damro, C., and Eckert, S. 2024. The geoeconomic turn of the single European market? Conceptual challenges and empirical trends. *JCMS: Journal of Common Market Studies*. doi: https://doi.org/10.1111/jcms.13587.

Hess, M., and R. Horner. 2024. Driving change in troubling times: Security risk and the state in global production networks. *ZFW–Advances in Economic Geography* 68 (3–4): 145–50. doi:10.1515/zfw-2024-0085.

Horner, R. 2017. Beyond facilitator? State roles in global value chains and global production networks. *Geography Compass* 11 (2): art. e12307. doi:10.1111/gec3.12307.

Horner, R., and Nadvi, K. 2018. Global value chains and the rise of the Global South: Unpacking twenty-first century polycentric trade. *Global Networks* 18 (2): 207–37. doi:10.1111/glob.12180.

IEA. 2021. Tracking clean energy innovation: Focus on China. https://www.iea.org/reports/tracking-clean-energy-innovation-focus-on-china.

IEA. 2024. World energy investment 2024. https://www.iea.org/reports/world-energy-investment-2024.

Jessop, B. 2001. Institutional re(turns) and the strategic–relational approach. *Environment and Planning A* 33 (7): 1213–35. doi:10.1068/a32183.
Klagge, B., Walker, B., Kalvelage, L., and Greiner, C. 2025. Governance of future-making: Green hydrogen in Namibia and South Africa. *Geoforum* 161: art. 104244. doi:10.1016/j.geoforum.2025.104244.
Kalvelage, L., and Tups, G. 2024. Friendshoring in global production networks: State-orchestrated coupling amid geopolitical uncertainty. *ZFW–Advances in Economic Geography* 68 (3–4): 151–66. doi:10.1515/zfw-2024-0042.
Langguth, H. 2024. Navigating conflictual cooperation: Temporary power coalitions in the planning and approval of large-scale Chinese green technology projects in Eastern Germany. In *Conflicts in urban future-making: Governance, institutions, and transformative change*, edited by M. Grubbauer, A. Manganelli, L. Volont, and M. Arndt. Transcript Verlag.
Lema, R., Fu, X., and Rabellotti, R. 2020. Green windows of opportunity: Latecomer development in the age of transformation toward sustainability. *Industrial and Corporate Change* 29 (5): 1193–209. doi:10.1093/icc/dtaa044.
Lew, L. 2024. China asks its carmakers to keep key EV technology at home. *Bloomberg*, September 12. https://www.bloomberg.com/news/articles/2024-09-12/china-asks-its-carmakers-to-keep-key-ev-technology-at-home.
Li, J., and Gong, H. 2025. Ambivalence in co-shaping an emerging global production network: Institutional complexity and strategic responses of Chinese and European actors in the EV battery industry. *Journal of Economic Geography*. doi: 10.1093/jeg/lbaf060.
Li, M. 2020. The Belt and Road Initiative: Geo-economics and Indo-Pacific security competition. *International Affairs* 96 (1): 169–87. doi:10.1093/ia/iiz240.
Lim, K. F. 2025. The firm as a geoeconomic actor. *Environment and Planning A: Economy and Space* 57 (1): 138–43. doi:10.1177/0308518X241265289.
Liu, F. C., Simon, D. F., Sun, Y. T., and Cao, C. 2011. China's innovation policies: Evolution, institutional structure, and trajectory. *Research Policy* 40 (7): 917–31. doi:10.1016/j.respol.2011.05.005.
Liu, W., and Dicken, P. 2006. Transnational corporations and 'obligated embeddedness': Foreign direct investment in China's automobile industry. *Environment and Planning A* 38 (7): 1229–47. doi:10.1068/a37206.
Mallin, F., and Sidaway, J. D. 2024. Critical geoeconomics: A genealogy of writing politics, economy and space. *Transactions of the Institute of British Geographers* 49 (1): art. e12600. doi:10.1111/tran.12600.
Mayer, F. W., and Phillips, N. 2017. Outsourcing governance: States and the politics of a 'global value chain world'. *New Political Economy* 22 (2): 134–52. doi:10.1080/13563467.2016.1273341.
Meckling, J. 2025. The geoeconomic turn in decarbonization. *Nature* 645 (8082): 869–76. doi:10.1038/s41586-025-09416-x.
MIIT. 2024. Lithium-ion battery industry specification conditions (2024 Edition). https://wap.miit.gov.cn/cms_files/filemanager/1226211233/attach/20246/1b2c425b31524352a0ea0571b51f1d9b.pdf.
MoC. 2023. Supporting the healthy development of NEV trade cooperation. https://www.gov.cn/zhengce/zhengceku/202402/content_6931276.htm.
MoC. 2025. Announcement on the decision to implement export controls on items related to lithium batteries and artificial graphite anode materials. https://aqygzj.mofcom.gov.cn/qdml/art/2025/art_8723634bd6f648d795426a656c0d54a0.html.
Parodi, A. 2025. Chinese EV makers sell more plugin-hybrids in the EU to avoid tariffs, research firm says. *Reuters*, May 2. https://www.reuters.com/business/autos-transportation/chinese-ev-makers-sell-more-plugin-hybrids-eu-avoid-tariffs-research-firm-says-2025-05-01/.
Pavlínek, P. 2024. Geopolitical decoupling in global production networks. *Economic Geography* 100 (2): 138–69. doi:10.1080/00130095.2023.2281175.
Sebastian, G. 2021. In the driver's seat: China's electric vehicle makers target Europe. *Merics*, September 1. https://merics.org/en/report/drivers-seat-chinas-electric-vehicle-makers-target-europe.

Sebastian, G. 2023. Watts the plan, Europe—Chinese battery investments on or off? *Merics*, July 31. https://merics.org/en/comment/watts-plan-europe-chinese-battery-investments-or.
Sebastian, G., Goujon, R., and Meyer, A., 2024. Pole position: Chinese EV investments boom amid growing political backlash. *Rhodium Group*, February 29. https://rhg.com/research/pole-position-chinese-ev-investments-boom-amid-growing-political-backlash/.
Selwyn, B., Bernhold, C., and Leyden, D. 2025. The geopolitical underpinning of global value chains and production networks: US–China technological rivalry in a longer-range perspective. *Journal of Economic Geography*. doi: 10.1093/jeg/lbaf033.
Smith, A. 2015. The state, institutional frameworks and the dynamics of capital in global production networks. *Progress in Human Geography* 39 (3): 290–315. doi:10.1177/0309132513518292.
Sparke, M. 1998. From geopolitics to geoeconomics: Transnational state effects in the borderlands. *Geopolitics* 3 (2): 62–98. doi:10.1080/14650049808407619.
Sparke, M. 2018. Globalizing capitalism and the dialectics of geopolitics and geoeconomics. *Environment and Planning A: Economy and Space* 50 (2): 484–89. doi:10.1177/0308518X17735926.
Teixeira, T. 2024a. Variegated forms of corporate capture: The state, MNCs, and the dark side of strategic coupling. *Global Networks* 24 (1): art. e12433. doi:10.1111/glob.12433.
Teixeira, T. 2024b. State–firm nexus, indigenous coupling, and social downgrading in the aerospace global production network: A case study of Embraer, Brazil. *Economic Geography* 100 (4): 303–28. doi:10.1080/00130095.2024.2346526.
Teixeira, T. 2024c. State power, global production networks, and underdevelopment: Examining South Carolina's success in fostering strategic coupling. *ZFW–Advances in Economic Geography* 68 (3–4): 213–25. doi:10.1515/zfw-2024-0044.
Teixeira, T., and Bridge, G. 2024. Creating elite encounters: The 'campaign' as approach for interviewing corporate elites. *Environment and Planning A: Economy and Space* 56 (8): 2121–42. doi:10.1177/0308518X241261701.
Umar, I. A., and Yang, C. 2025. Geoeconomic bridging node in global production: Chinese electric vehicle battery investments in Morocco. *Contemporary Social Science*. doi: 10.1080/21582041.2025.2571579.
UN Environment Programme. 2019. The Belt and Road Initiative International Green Development Coalition (BRIGC). https://www.unep.org/regions/asia-and-pacific/regional-initiatives/belt-and-road-initiative-international-green.
Vezzoni, R. 2024. A 'return of the state' in energy transitions? The making of a hydrogen economy in the European Union. *ZFW–Advances in Economic Geography* 68 (3–4): 195–212. doi:10.1515/zfw-2024-0050.
Werner, M. 2021. Geographies of production II: Thinking through the state. *Progress in Human Geography* 45 (1): 178–89. doi:10.1177/0309132520911996.
Wiggins, K., Li, G., McMorrow, R., Dempsey, H., and Alim, A. N. 2024. China's battery giant taps Europe's elite to expand supply chain. *Financial Times*, July 12. https://www.ft.com/content/aaf15b05-0aa0-49d6-a8d8-9b162fdd6771.
Xie, S. Y. 2024. Chinese EV battery maker SVOLT to shut European operations. *Nikkei*, October 25. https://asia.nikkei.com/Business/Technology/Chinese-EV-battery-maker-SVOLT-to-shut-European-operations.
Yan, Z., Li, Q., Wang, E., and Lee, L. 2024. China to invest more than $830 mln in solid-state battery research -source. *Reuters*, May, 29. https://www.reuters.com/world/china/china-invest-more-than-830-mln-solid-state-battery-research-source-2024-05-29/.
Yeung, H. W.-c. 2023. Troubling economic geography: New directions in the post-pandemic world. *Transactions of the Institute of British Geographers* 48 (4): 672–80. doi:10.1111/tran.12633.
Zhang, J. 2024. How to better go abroad in the face of trade barriers. *China Automotive News*, September 24. http://www.cnautonews.com/shendu/2024/09/24/detail_20240924368335.html.
Zhang, Y. 2025. China's NEV production, sales surpass 12 m each, leading globally in 2024 for 10th year. *Global Times*, January 13. https://www.globaltimes.cn/page/202501/1326817.shtml.